

SECTION B

Answer all questions in the spaces provided.

7. (a) Ethanol, $\text{C}_2\text{H}_5\text{OH}$, is a liquid at room temperature. Write the equation that corresponds to the standard enthalpy change of formation of ethanol. [2]

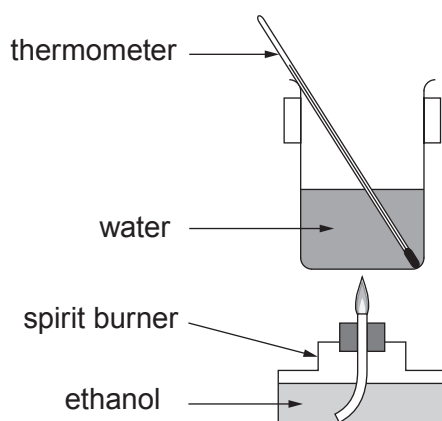
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- (b) Why is it difficult to measure the standard enthalpy change of formation of ethanol directly? [1]

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- (c) The following apparatus can be used to determine the enthalpy change of combustion of ethanol, $\Delta_c H$.



- (i) Explain how the enthalpy change of combustion is calculated from the results collected using this apparatus. [3]

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- (ii) Why is it difficult to use this apparatus to determine the enthalpy change of combustion of ethane, C_2H_6 ? [1]

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- (d) Two students carried out an experiment using the apparatus in part (c).

Neither followed the given method correctly. Euan extinguished the flame after only 10 seconds. Carys continued to heat for two minutes after the water had boiled.

Explain the effect that each of the students' errors had on the values they obtained in this experiment. [2]

Effect of Euan's error

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Effect of Carys' error

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- (e) Standard enthalpy changes of formation, $\Delta_f H^\theta$, can be found by using standard enthalpy changes of combustion, $\Delta_c H^\theta$.

The table shows some standard enthalpy changes of combustion.

Substance	Standard enthalpy change of combustion / kJ mol^{-1}
carbon, C	–394
hydrogen, H_2	–286
ethanol, $\text{C}_2\text{H}_5\text{OH}$	–1371

- (i) Use these data to calculate the standard enthalpy change of formation of ethanol. Show clearly how you carried out your calculation. [3]

$$\Delta_f H^\theta = \dots\dots\dots \text{kJ mol}^{-1}$$

- (ii) Amir said that hydrogen is the best fuel of these three in terms of the energy released by burning equal masses. Is he correct? Justify your answer. [2]

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